V2AP: Learning Affordance Posteriors for Robotic Grasping from Human Video Priors

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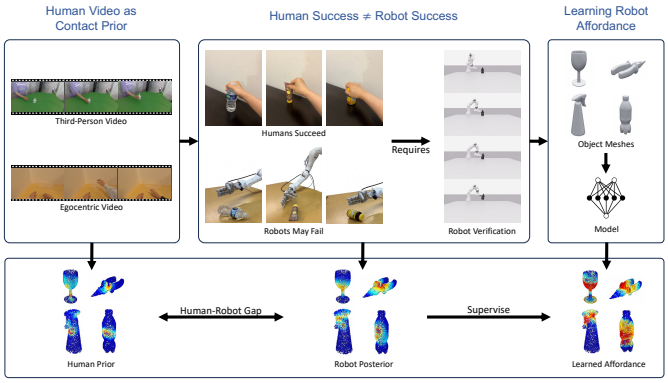


Fig. 1. **Data-centric overview of V2AP.** Human videos provide scalable object-centric contact priors, but raw human contacts are not directly robot-executable. V2AP grounds these priors through robot verification to construct Robot Posteriors, which supervise robot affordance learning.

***Abstract*—Human videos are a scalable source of manipulation data, but they are not directly robot-executable: human contacts are embodiment-agnostic, lack action labels, and often violate robot reachability, collision, or grasp-stability constraints. This paper studies how such abundant but weakly structured human video data can be converted into robot-usable supervision. We present V2AP, a data-centric framework that first extracts object-centric Human Priors from egocentric and third-person human videos, and then filters these priors through robot embodiment and physics to construct Robot Posteriors. Candidate grasps are generated from high-prior object regions and verified using collision-aware motion planning and closed-loop simulation. Successful rollouts are aggregated into dense posterior affordance maps, which serve as robot-executable labels for learning affordance predictors and grasp generators. Experiments show that training on Robot Posteriors improves grasp success over directly using raw Human Priors, suggesting that human videos are most useful for robot learning when treated as scalable priors rather than direct demonstrations. Our results support affordance posterior learning as a practical data-curation mechanism for transforming human video priors into physically grounded robot manipulation data.**

I. Introduction

Robotic manipulation has achieved remarkable progress with large-scale robot datasets and imitation learning [12, 25, 38, 32, 23]. However, collecting diverse robot manipulation data remains fundamentally expensive and difficult to scale, especially for long-horizon or contact-rich tasks [15, 4, 18]. In contrast, the internet contains massive amounts of human interaction videos that capture rich object-centric manipulation knowledge across diverse objects, tasks, and environments. These videos implicitly encode how humans interact with objects, revealing affordances such as where to grasp, push, or pull to accomplish manipulation goals. From a data-centric perspective, the key challenge is not only how to learn from human videos, but how to transform them into supervision that is compatible with robot embodiment and physical execution.

Despite this potential, directly transferring manipulation knowledge from human videos to robots remains challenging [34, 38, 26]. Human demonstrations are unconstrained by robot embodiment, kinematics, or physical limitations, and many observed interactions are not directly executable for robots. Moreover, human videos typically lack action labels, calibrated geometry, and explicit physical supervision, making it difficult to extract robot-consistent manipulation trajectories. Existing approaches often rely on expensive teleoperation [34, 26, 35], manually collected demonstrations [17, 13, 6], or direct video imitation, which is difficult to scale and may struggle to generalize across diverse objects and environments.

This raises a key question: what should be transferred from human videos to robots? Directly transferring human motion is brittle because motion trajectories are embodiment-specific, while transferring only high-level task semantics discards the contact-level information required for manipulation. We argue that *affordance* is the appropriate mid-level representation. Affordances encode object-centric interaction possibilities, such as where an object can be grasped and which surface regions are relevant for physical interaction. They are easier to infer from human videos than full robot trajectories, yet more informative for manipulation than task-level labels.

Crucially, however, affordances extracted from human videos are still only priors. A human contact prior indicates where humans tend to interact with an object, but it does not answer whether a particular robot can successfully interact with the same region. A visually plausible contact may not admit a valid gripper approach, may collide with the object or scene, may violate robot reachability, or may fail to produce a stable grasp. Therefore, human affordance priors must be grounded in object geometry and filtered through robot embodiment and physical constraints before they can serve as robot supervision. This motivates our formulation of *affordance posterior learning*: converting broad human interaction priors into robot-executable affordance posteriors.

To this end, we introduce V2AP, a framework that learns robotic affordance posteriors from unlabeled human videos. V2AP first reconstructs hand-object interactions from egocentric and third-person videos and aggregates contact evi-

dence on the object surface to obtain a dense Human Prior. It then generates human-prior-guided robot grasp candidates and evaluates them in simulation under inverse kinematics, collision constraints, gripper geometry, reachability, and grasp stability. Only successful rollouts are retained to form the Robot Posterior, which answers a different question from the Human Prior: not simply "where do humans grasp?", but "among human-plausible regions, where can the robot successfully grasp?"

Finally, V2AP uses the Robot Posterior as supervision to train a dense affordance predictor and an affordance-conditioned diffusion model for grasp pose generation. At inference time, the predicted posterior affordance map guides pose sampling toward regions that are both human-plausible and robot-feasible. Overall, our work frames human-video-to-robot transfer as a data construction problem. Rather than treating human videos as direct demonstrations, V2AP treats them as scalable priors that must be verified, filtered, and relabeled into robot-executable posteriors.

II. Related Work

Human video priors and affordance learning. Human videos provide scalable cues for manipulation by revealing object use, contact patterns, and interaction intent across diverse scenes. Prior works use large human-video corpora to learn transferable robot representations or rewards [22, 19], while VRB extracts visual affordances from human interaction videos for robotic manipulation [1]. In parallel, affordance learning predicts actionable regions from images or 3D observations, from AffordanceNet to interaction-centric 3D methods such as Where2Act [8, 21]. However, these methods either use human videos as generic priors or learn affordances without explicitly resolving robot embodiment mismatch. We instead treat human-observed interactions as object-centric priors and infer robot-executable affordance posteriors conditioned on embodiment and physical feasibility.

Embodiment-grounded transfer and executable manipulation. Human-to-robot transfer often reconstructs hand-object motion with hand models or monocular recovery and retargets it to robot demonstrations [28, 24, 27]. Complementary grasping and visuomotor methods learn executable contacts, 6-DoF grasps, or action distributions from robot-centric data, including Dex-Net, GraspNet, Contact-GraspNet, Diffusion Policy, and DP3 [20, 9, 30, 5, 36]. Our use of retargeting and simulation is different: they serve as closed-loop physical verification rather than final supervision. By filtering broad human priors through reachability, collision, and grasp-stability constraints, we turn embodiment and physics into a posterior-construction step for affordance learning.

III. Constructing Robot-Executable Data from Human Video Priors

Our key insight is to convert human contact priors into robot-executable posteriors. Human videos provide object-centric contact priors that reveal where people naturally touch an object, but these regions are not necessarily feasible for a robot due to embodiment mismatch, reachability, collision constraints, and grasp stability. We therefore use simulation and cuRobo-based filtering to transform the Human Prior into a Robot Posterior. This posterior preserves human grasping awareness while grounding it in robot-specific physical feasibility.

A. Data Formulation: Human Priors and Robot Posteriors

We use multiple egocentric and third-person human-object interaction video datasets to learn where a robot can successfully grasp an object. For each object, we reconstruct a mesh and sample a surface point cloud $\mathcal{P} = \{p_i\}_{i=1}^{N}$, where $p_i \in \mathbb{R}^3$ is a 3D surface point.

Our method uses three related but distinct heatmaps on the object surface. The **Human Prior** $\mathbf{h} = \{h_i\}_{i=1}^{N}$ is extracted from human hand-object contact in videos and indicates where humans tend to touch the object. The **Robot Posterior** $\mathbf{y} = \{y_i\}_{i=1}^{N}$ is generated through prior-guided grasp verification in simulation and indicates which contact regions are physically feasible for the robot. The **Predicted Affordance** $\hat{\mathbf{y}} = \{\hat{y}_i\}_{i=1}^{N}$ is produced by our learned model at test time and serves as the affordance signal for grasp generation.

The objective of this paper is to train a model that predicts robot-executable affordances from object point clouds:

$$\hat{\mathbf{y}} = f_\theta(\mathcal{P}), \qquad \hat{y}_i \in [0, 1]. \tag{1}$$

In this process, $\mathbf{h}$ guides data generation, and $\mathbf{y}$ provides the training target. This design converts human videos into meaningful robot learning supervision. During deployment, the model does not require human videos, hand poses, simulation labels, or full action trajectories. By learning an object-centric affordance representation rather than an end-to-end policy, our framework enables efficient and readily deployable robot grasping from observed object geometry.

B. Extracting Human Contact Priors from Videos

Dual-View Human Video Processing. We extract human contact evidence from both egocentric and third-person videos. For egocentric videos, we use MegaSAM [16] and HaWoR [37] to estimate camera information, scene geometry, and hand parameters. For third-person videos, we use DepthPro [2] and HaPTIC to obtain corresponding camera, depth, and hand motion information. In both cases, we reconstruct the target object mesh using SAM3D [31] and estimate its 6-DoF pose over time using FoundationPose.

Hand-Object Alignment. For each video frame t, we reconstruct the MANO [29] hand mesh $\mathcal{H}_t$ and align it with the object mesh in a shared world coordinate frame. This alignment allows hand-object contacts from different frames, videos, and viewpoints to be accumulated on the same object surface.

Contact Aggregation. For each object surface point p_i, we compute its closest distance to the hand mesh. A point is marked as contacted if this distance is below a threshold $\tau_c = 3\,\mathrm{cm}$. We aggregate contact evidence over all videos and frames containing the same object instance $C_i = \sum_{v,t} c_{i,t}$.

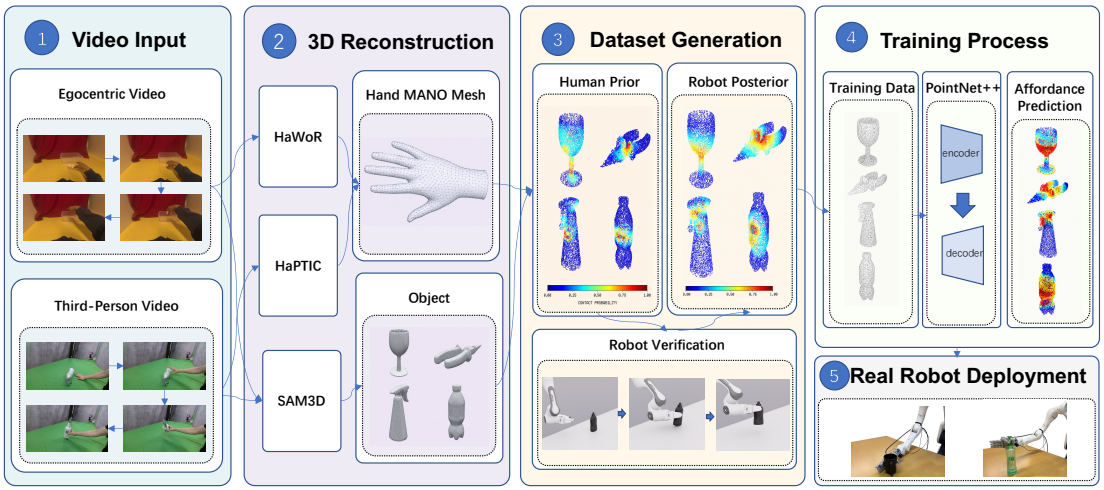


Fig. 2. **Data construction pipeline of V2AP.** With egocentric and third-person videos, we reconstruct the MANO hand mesh and object mesh, aggregate human-object contacts into a Human Prior Map, and use this prior to generate robot grasp candidates. Physics-based verification converts these candidates into a Robot Posterior Map, which supervises a PointNet++ affordance predictor for novel objects and real-world deployment.

The resulting raw contact count C_i gives sparse evidence of where humans physically touch the object.

Human Prior. To obtain a continuous Human Prior heatmap, we diffuse the sparse contact counts over the object surface with a Gaussian kernel. After diffusion and normalization, we obtain the Human Prior $\mathbf{h} = \{h_i\}_{i=1}^{N}$, where $h_i \in [0, 1]$. This heatmap captures human-preferred contact regions. However, human contact does not necessarily satisfy robot gripper geometry, reachability, or grasp stability constraints, so this prior must be further converted into robot-executable supervision.

C. Robot-Centric Filtering via Simulation Verification

Prior-Guided Grasp Candidate Generation. We convert the Human Prior into robot-executable supervision by using $\mathbf{h}$ to bias parallel-jaw grasp sampling toward high-prior contact regions, instead of directly imitating human wrist poses. A primary contact point c_1 is sampled from points with $h_i > 0.3$ with probability proportional to h_i, and an opposing contact point c_2 is searched by casting rays along 32 tangent-plane directions of c_1. We retain only contact pairs that satisfy the Franka gripper width range, are approximately antipodal, avoid invalid penetration, remain close to high-confidence human-prior regions, and do not induce table-colliding bottom-up approaches.

Simulation-Based Verification. The remaining grasp candidates are imported into Isaac Sim and verified in two stages. First, we use cuRobo as a collision-aware kinematic feasibility filter. A candidate is retained only if cuRobo can find a valid, collision-free robot motion to the target grasp pose. Candidates that are unreachable, collide with the object or environment, or fail motion planning are discarded. Second, each feasible grasp is executed in Isaac Sim, where it is labeled successful only if the robot closes the gripper on the object and stably lifts it.

For each successful grasp, we record the final gripper pose in the object coordinate frame. Using the known gripper geometry, we back out the contact points. These inferred successful contacts are collected into a set $\mathcal{C}^*$. A binary robot contact label is then assigned to each surface point by nearest-neighbor lookup.

Robot Posterior. We convert sparse binary contact labels ℓ_i into a continuous Robot Posterior heatmap $\mathbf{y}$ by applying Gaussian diffusion on the object surface. Compared with the Human Prior $\mathbf{h}$, $\mathbf{y}$ represents an embodiment-conditioned estimate of where successful robot grasps are physically attainable.

D. Posterior-Supervised Affordance Learning

After generating the Robot Posterior Map $\mathbf{y}$, we train a PointNet++-based affordance network to predict robot-executable grasp regions directly from object geometry. Given an object point cloud $\mathcal{P} = \{p_i\}_{i=1}^{N} \in \mathbb{R}^{N\times 3}$, the network outputs a dense Predicted Affordance Map $\hat{\mathbf{y}} = f_\theta(\mathcal{P})$, where each $\hat{y}_i$ estimates the posterior grasp affordance of surface point p_i. We supervise this prediction using the simulation-generated Robot Posterior $\mathbf{y}$ as the target. Since $\mathbf{y}$ is a continuous posterior map, we use a point-wise ℓ_1 loss over all object surface points:

$$\mathcal{L}_{\text{aff}} = \frac{1}{N}\sum_{i=1}^{N} |\hat{y}_i - y_i| \,. \tag{2}$$

This encourages the network to preserve the graded confidence structure of physically attainable grasp regions.

E. Affordance-Conditioned Grasp Pose Generation

The predicted posterior affordance map indicates where the robot should grasp, but execution requires a full 6-DoF end-effector pose. We therefore train a conditional diffusion model

to generate grasp poses $g = (t, R)$, where t is the gripper position and R is the gripper orientation, conditioned on the object point cloud $\mathcal{P}$ and predicted affordance map $\hat{\mathbf{y}}$. The model learns the conditional distribution $p_\psi(g \mid \mathcal{P}, \hat{\mathbf{y}})$, where $\hat{\mathbf{y}}$ biases generation toward high-confidence robot-executable regions.

Following standard denoising diffusion training, we optimize

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{g_0,k,\epsilon}\left[\left\|\epsilon - \epsilon_\psi\left(g_k, k, \phi(\mathcal{P}, \hat{\mathbf{y}})\right)\right\|_2^2\right], \tag{3}$$

where g_0 is a successful grasp pose, g_k is its noised version at diffusion step k, ϵ is Gaussian noise, and $\phi(\mathcal{P}, \hat{\mathbf{y}})$ encodes the affordance-conditioned point cloud.

At inference time, we sample multiple grasp candidates and apply a two-stage selection procedure. First, we use a hard geometric gate to remove invalid poses, including grasps whose gripper penetrates the object or whose wrist-frame orientation is flipped with respect to the executable approach direction. We then filter the remaining candidates using inverse-kinematics feasibility and collision checking, and select the feasible grasp with the highest affordance score. The selected grasp is passed to the robot motion planner for execution.

IV. Experiments

We design experiments to evaluate whether human video priors can be converted into robot-executable affordance posteriors. Specifically, our experiments aim to answer three questions:

- Can our framework capture meaningful grasp regions from Human Priors learned from egocentric and third-person human videos?
- How much improvement can robot-feasible Robot Posteriors bring compared with raw Human Priors?
- How does posterior-supervised affordance learning compare with raw human-prior supervision and representative grasping baselines?

A. Dataset and Metrics

Dataset. We use human grasping videos from three open-source datasets: OakInk [33], DexYCB [3], and EgoDex [14]. These datasets provide diverse human-object interactions with different object categories, hand poses, camera viewpoints, and grasp strategies. The objects are split into training and test objects to evaluate generalization to unseen object instances, denoted by Unseen Objs in Table I. Furthermore, we gather objects from unseen categories, denoted by Unseen Cats in Table I, to evaluate category-level generalization.

Metric. We use Grasp Success Rate (GSR) as the metric. We conduct grasp prediction on each test object and report the success rates on both unseen objects and unseen categories.

B. Baselines and Ablations

To contextualize the effectiveness of posterior-supervised affordance learning, we compare it with the following representative baselines:

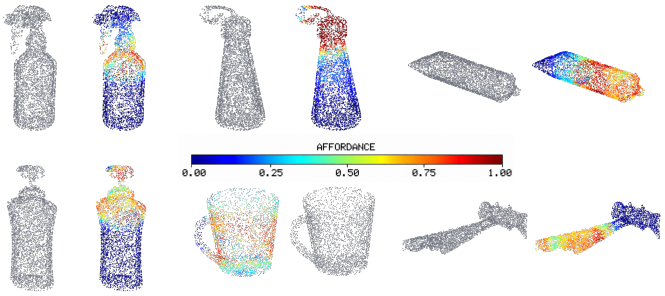


Fig. 3. **Visualization of Robot Posterior Affordance Maps.** This figure visualizes the raw point cloud and the predicted affordance of objects from six different categories: spray bottle, dispenser, bottle, mug, toothpaste, and hammer.

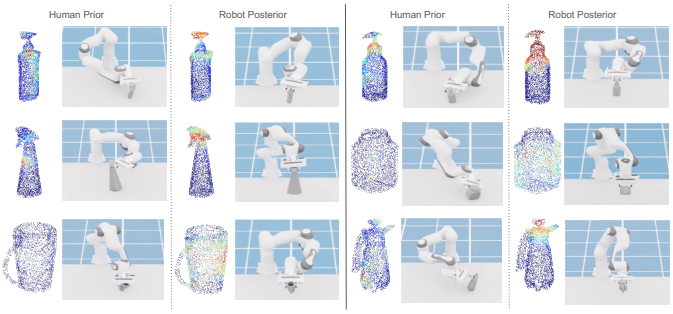


Fig. 4. **Comparisons Between Grasping Guided by Human Prior and Robot Posterior.** Raw Human Priors may lead to robot-infeasible grasps, including collisions and unstable contacts, while Robot Posteriors bias grasping toward physically verified regions.

- **3D Diffusion Policy (DP3)** [5, 36]: a diffusion-based method with a 3D backbone that imitates human grasps in videos.
- **GraspVLA** [7]: a VLA method that predicts grasps trained on large-scale simulation data.
- **GraspNet** [10, 11]: an end-to-end method trained on large-scale real-world data with different objects and scenes.

To isolate the value of robot-centric data filtering, we compare against the following ablation:

- **Ours w/o P(osterior)**: our method without Robot Posterior, *i.e.*, the affordance predictor is trained on the Human Prior.

C. Quantitative and Qualitative Evaluations

Figure 3 visualizes the predicted Robot Posterior affordance maps across diverse object categories. The highlighted areas denote points predicted to be suitable for robot grasping. These visualizations suggest that the learned posterior maps capture consistent robot-executable grasp regions across diverse object categories and shapes.

Table I shows that our posterior-supervised model achieves higher grasp success rates than the compared baselines in our evaluation setting. The advantage of our method mainly comes from its object-centric affordance representation. This representation mitigates the influence of variation in robot spatial

TABLE I
GRASP SUCCESS RATE ON UNSEEN OBJECTS IN SEEN CATEGORIES AND UNSEEN CATEGORIES. THE BEST METHOD FOR EACH EVALUATION CATEGORY IS HIGHLIGHTED IN **BOLD**.

Success Rate (%)	Ours	Ours w/o P	GraspNet	DP3	GraspVLA
Unseen Objs	**63.1**	35.2	34.5	25.1	9.2
Unseen Cats	**57.0**	22.1	31.0	13.1	4.1
Average	**61.6**	31.9	33.6	24.2	7.9

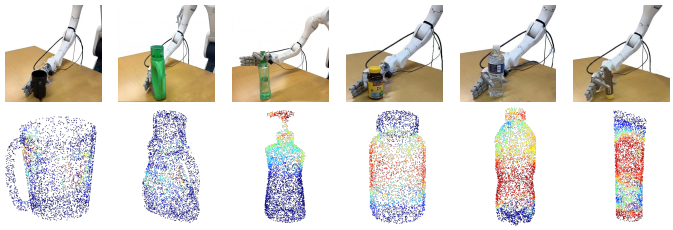


Fig. 5. **Real-World Affordance and Grasping.** We selected six real-world objects to provide qualitative evidence of real-world transfer. The robot obtains object meshes and point clouds from a single RGB-D image, predicts posterior affordance maps, and executes generated grasp poses.

trajectories, object orientations, and poses, allowing the policy to focus more directly on object properties for grasping, such as geometry and local interaction structure. Furthermore, our method exhibits a relatively small performance gap between unseen objects and unseen categories, suggesting that point-level posterior affordances can generalize to objects from novel categories with similar local grasping cues.

The gap between Ours and Ours w/o Posterior highlights the importance of converting human priors into robot-executable labels. Raw Human Priors identify where humans tend to contact objects, but these regions may be infeasible for a robot due to gripper geometry, reachability, collision, or unstable contact. In contrast, Robot Posteriors are constructed from verified successful rollouts, providing a curated supervision signal that better matches robot execution constraints. Figure 4 further illustrates this distinction: grasping guided directly by Human Priors may lead to collisions or unstable contact, while grasping guided by Robot Posteriors is biased toward physically verified contact regions.

D. Real-Robot Evaluation

We evaluate the full pipeline on a physical Dexmate robot with a Sharpa two-finger hand to examine whether the learned Robot Posterior affordances transfer from simulation to real-world execution. At test time, the robot observes the object with an RGB-D camera, constructs the mesh and 3D point cloud as in our framework, predicts the Robot Posterior affordance map, and generates grasp poses with the affordance-conditioned diffusion model. Figure 5 shows qualitative examples of predicted affordances and corresponding real-world grasping behaviors.

V. CONCLUSION

We presented V2AP, a data-centric framework for converting human video priors into robot-executable affordance posteriors. Instead of treating human videos as direct robot demonstrations, V2AP extracts object-centric Human Priors and filters them through robot embodiment and physical verification to construct Robot Posteriors. These posteriors provide structured supervision for learning affordance maps and grasp generators from object geometry. Our results suggest that the value of human video data for robot learning lies not only in scale, but also in how the data is curated, grounded, and relabeled into robot-compatible supervision.

Limitations and Future Work. Our current framework focuses mainly on single-object grasping. Extending posterior construction to bimanual, multi-object, and long-horizon manipulation will require handling more severe occlusion, richer contact dynamics, and coordination constraints. Future work will also study failure-driven data flywheels, where robot failures are fed back into the posterior construction process to continuously improve the quality of robot-executable supervision.

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